

JING-ZHANG RESONANCE

CENTENNIAL JING-ZHANG AI INNOVATION BELT

JING-ZHANG RESONANCE

TUNING RAIL, BELLS, ALGORITHMS AND EVERYDAY LIFE

The city does not obey an algorithm; technology must listen to people. AI is treated as infrastructure with bodies, watts, latency, failure modes and public responsibility.

01 ONE RESONANCE SPINE

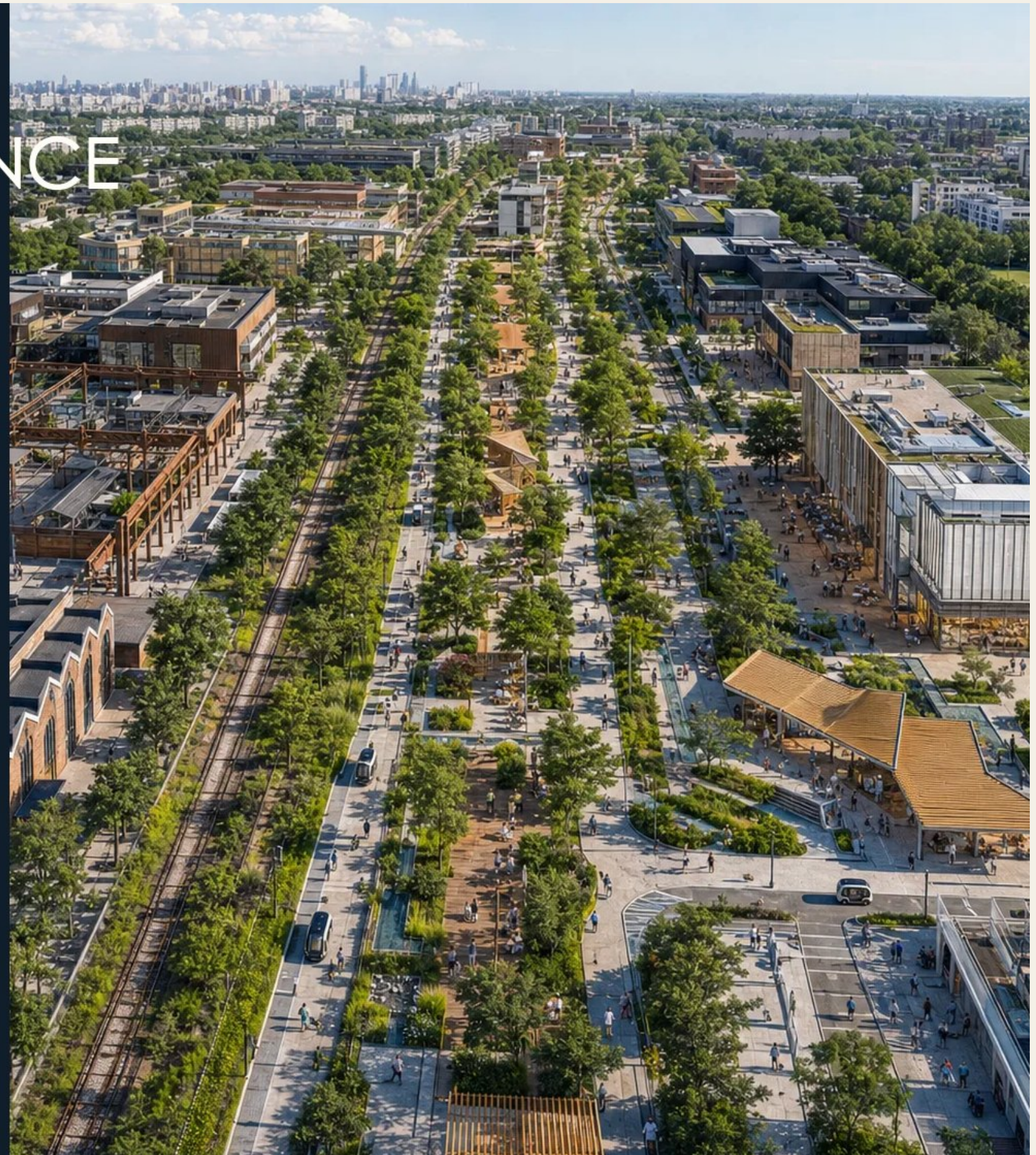
02 THREE FUTURE WORKYARDS

03 SIX CIVIC SYSTEMS

04 SIXTEEN EVERYDAY PROTOTYPES

Local DNA: Jing-Zhang engineering · Qinghuayuan inquiry ·
Dazhongsi bell rhythm · Haidian green corridors and everyday life

PROVISIONAL ROUGH BOUNDARY · NOT AN OFFICIAL REDLINE · CONCEPT
ONLY · RECALCULATE THE WHOLE SYSTEM WHEN OFFICIAL GEOMETRY
ARRIVES

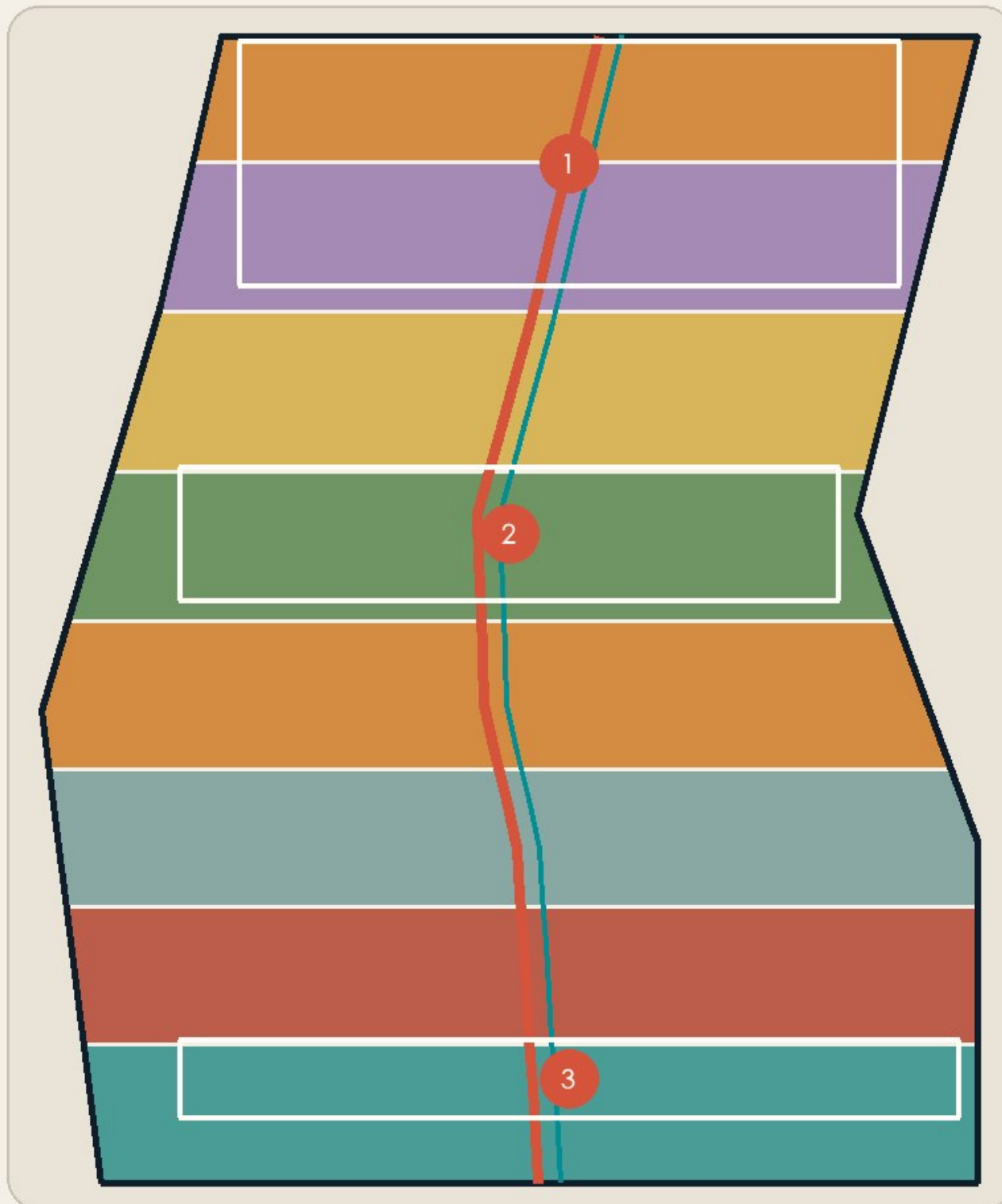


ONE SPINE, FOUR LOCAL ECHOES

RAIL / BELL / CAMPUS / GREEN CORRIDOR

ONE SPINE, FOUR LOCAL ECHOES

02



LOCAL CULTURE → SPATIAL MOVE

- **Jing-Zhang: self-reliant design, construction and operation**
 Turn rail heritage into a walkable, learnable, testable public spine
- **Qinghuayuan station: rail meets scholarship**
 Link campus, community and R&D through an open semantic twin
- **Dazhongsi: bells, time and resonance**
 Write visible rules, human review and refusal into AI services
- **Haidian green corridors: culture, ecology and daily life**
 Organize continuous life with canopy, rainwater, quiet lanes and ground-floor interfaces

SPATIAL STRUCTURE

One Centennial Resonance Spine / three future workyards / two collaborative wings / six civic systems / sixteen everyday prototypes

THREE FUTURE WORKYARDS

FOUND / LEARN / ADOPT

03

THREE FUTURE WORKYARDS, THREE MATURITIES

ZHONGZHIYUAN · ENGINEERING VOICE

PHYSICAL AI FOUNDRY



PUBLIC EDGE / SEMI-PUBLIC COURT / BOUNDED CORE

01

Separate training loop from public route

02

Sim → real → retrospective

03

Human takeover and physical stop

SECTION CONTROL

Open ground floor; research/learning above; retain egress, fire, accessibility and human service window.

AI ORIGIN · INQUIRY VOICE

CIVIC INTELLIGENCE ACADEMY



PUBLIC EDGE / SEMI-PUBLIC COURT / BOUNDED CORE

01

Public model room

02

Civic AI assembly

03

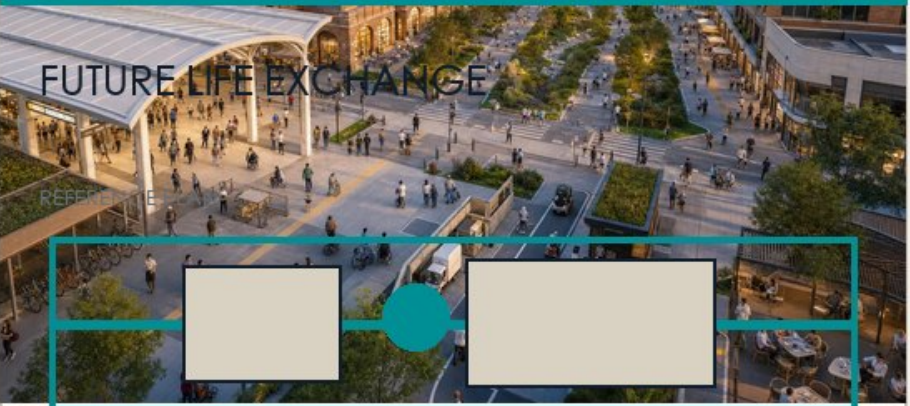
Human service court

SECTION CONTROL

Open ground floor; research/learning above; retain egress, fire, accessibility and human service window.

DAZHONGSI · BELL-RHYTHM VOICE

FUTURE LIFE EXCHANGE



PUBLIC EDGE / SEMI-PUBLIC COURT / BOUNDED CORE

01

Four-quadrant ground reconnection

02

Assistive technology library

03

Adoption clinic and refusal

SECTION CONTROL

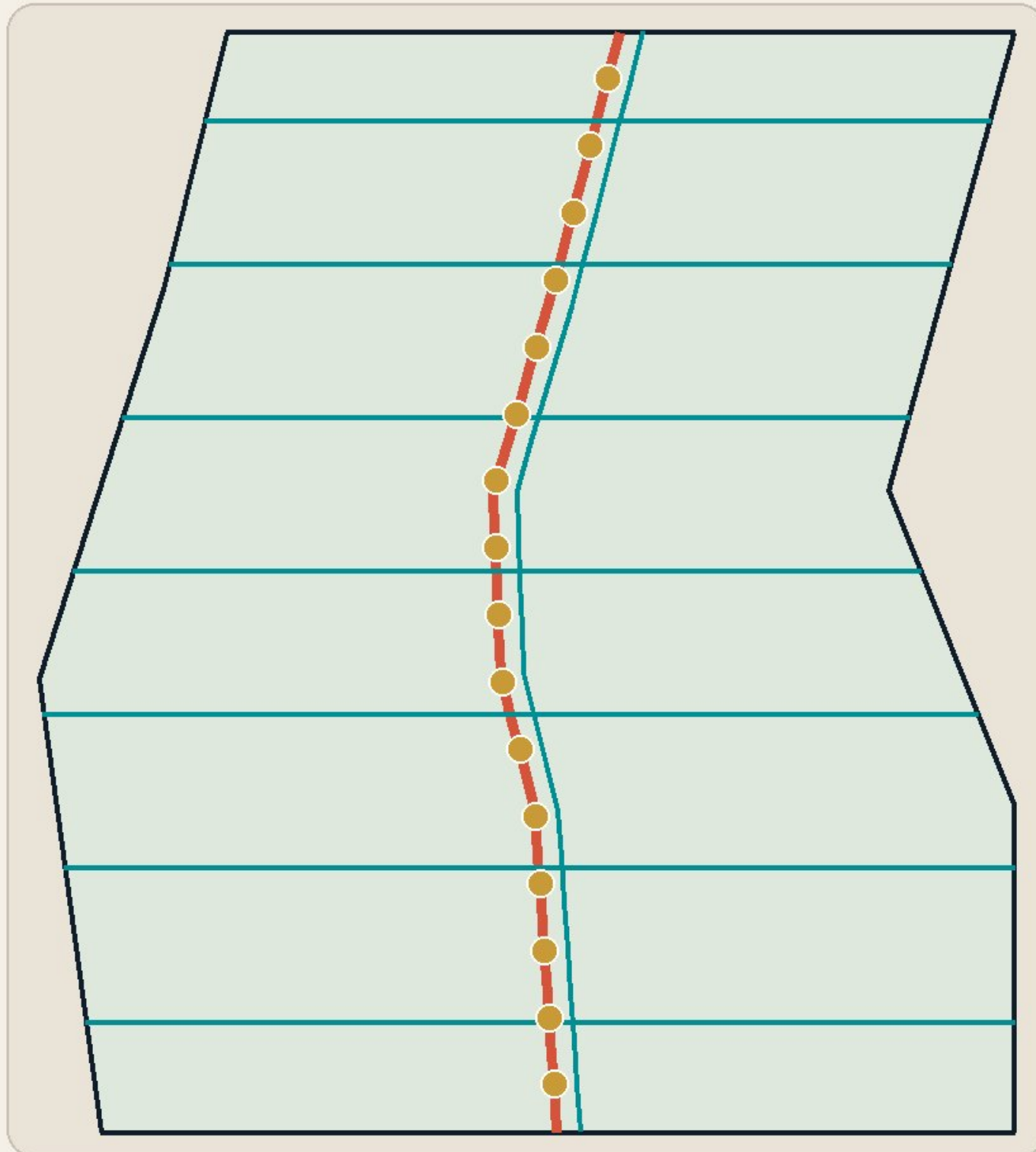
Open ground floor; research/learning above; retain egress, fire, accessibility and human service window.

PEOPLE FIRST; MACHINES LEARN TO WAIT

ACCESS / SEPARATION / CLIMATE

PEOPLE FIRST; MACHINES LEARN TO WAIT

04



STREET SECTION + HUMAN DETAIL



REFERENCE SECTION: QUIET WALK / CYCLE / LOW-SPEED SERVICE

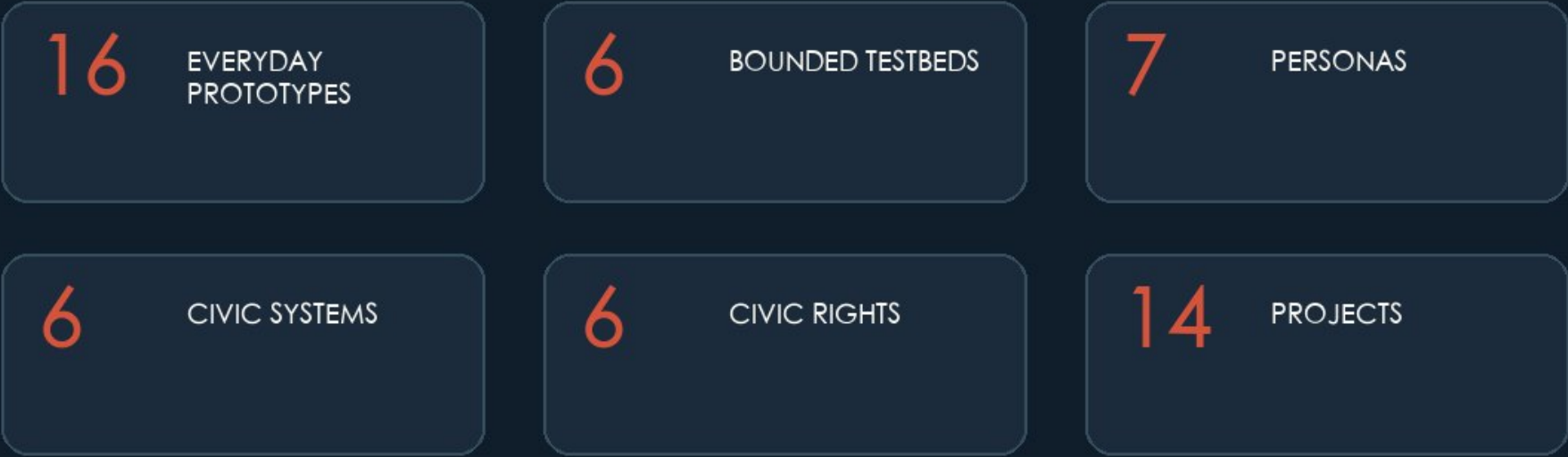


Hard edge: physical yield line, stop post, human service point. Soft edge: shade, rainwater, seating and low-

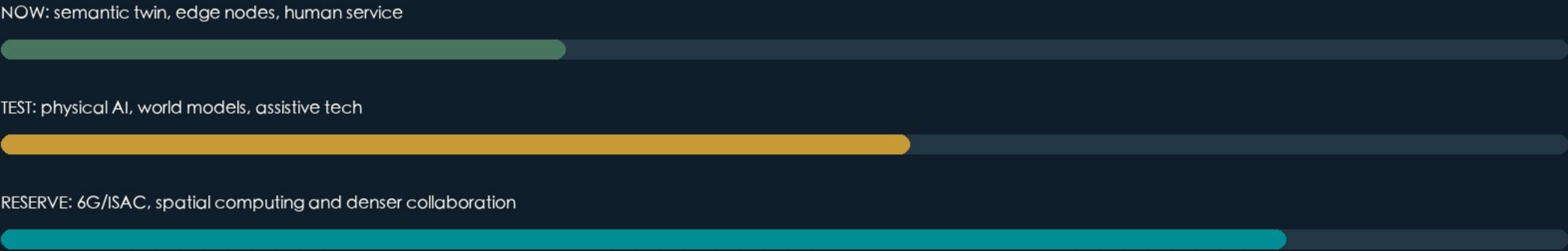
METRICS ARE GATES, NOT SLOGANS

MEASURE WHAT PEOPLE CAN VERIFY

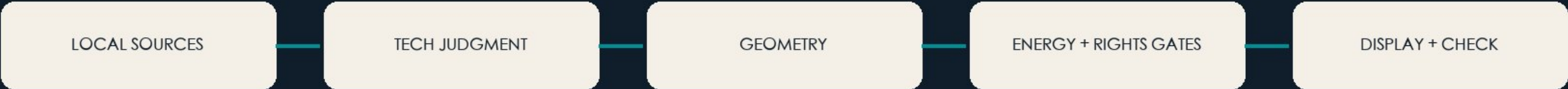
METRICS ARE GATES, NOT SLOGANS



TECHNOLOGY MATURITY + SPATIAL CARRIERS



AUDIT CHAIN



SIX CIVIC SYSTEMS

DATA / BODY / EDGE / ENERGY / MOBILITY / RIGHTS

06

SIX CIVIC SYSTEMS: TECHNOLOGY MUST BECOME FORM

L1	Open semantic city twin 现在 / NOW	City data rooms and public model lab	Spatial rule: disconnectable, observable, maintainable, human-takeover ready
L2	Physical AI and world models 试验 / TEST	Training loop, simulation rooms and depot	Spatial rule: disconnectable, observable, maintainable, human-takeover ready
L3	Edge compute and 6G-ready reserve 预留 / RESERVE	Disconnectable edge nodes and utility reserve	Spatial rule: disconnectable, observable, maintainable, human-takeover ready
L4	Compute-energy-water loop 逐址 / SITE TEST	Heat pumps, thermal store and rain gardens	Spatial rule: disconnectable, observable, maintainable, human-takeover ready
L5	Autonomous and accessible mobility 受控 / BOUNDED	Quiet walk, cycle and machine-service lanes	Spatial rule: disconnectable, observable, maintainable, human-takeover ready
L6	Civic AI governance and human fallback 先行 / FIRST	Human service court, audit desks and appeals	Spatial rule: disconnectable, observable, maintainable, human-takeover ready

6G is a reserve; heat reuse is site-specific feasibility; world models start in simulation and bounded training.

THE CITY NEEDS A HUMAN OFF SWITCH

THE CITY HAS A HUMAN OFF SWITCH

SIX NON-DEROGABLE CIVIC AI RIGHTS

- 01

Right to know

Applies to every scenario by default
- 02

Refuse without penalty

Applies to every scenario by default
- 03

Equivalent human service

Applies to every scenario by default
- 04

Privacy and minimum data

Applies to every scenario by default
- 05

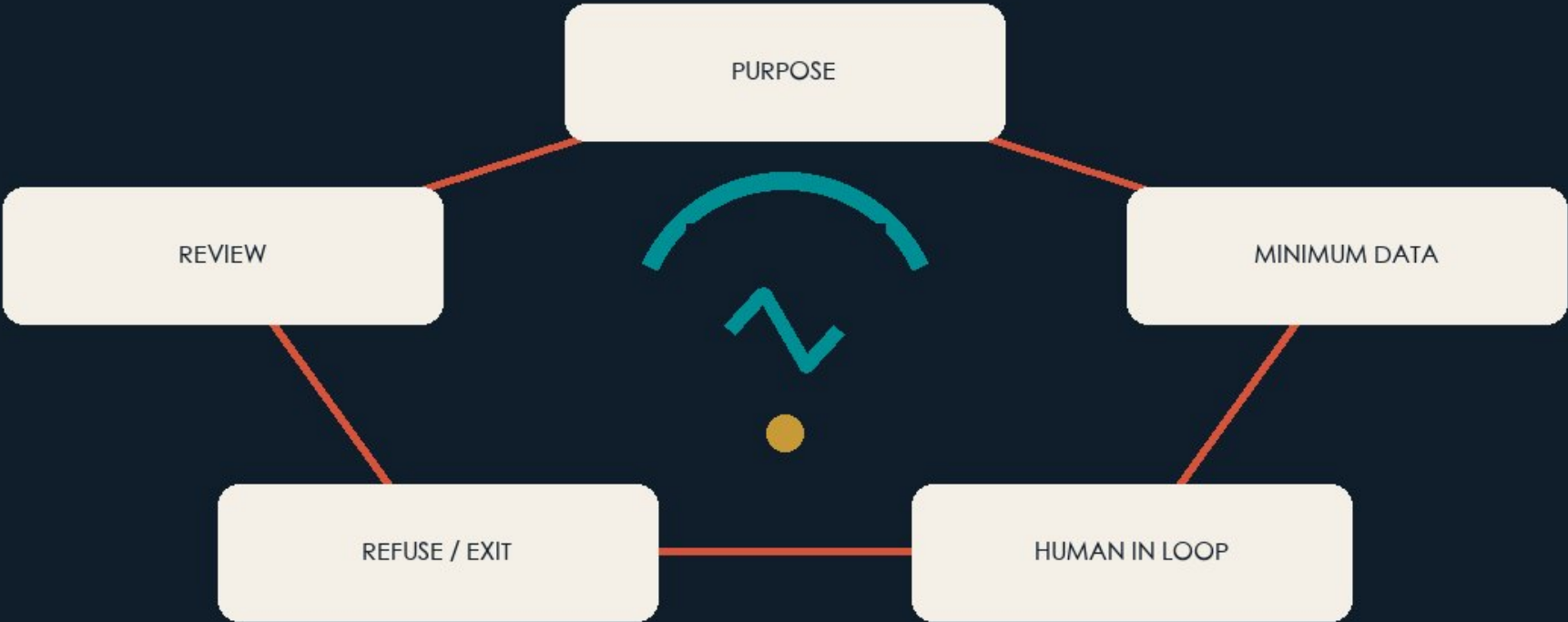
Appeal and human review

Applies to every scenario by default
- 06

Know energy and environmental cost

Applies to every scenario by default

FIVE-STEP GOVERNANCE LOOP



No public pilot without equivalent human service, physical off, an appeal route and energy disclosure.

A CITY SECTION FROM TRAINING TO ADOPTION

AERIAL / EYE LEVEL / DETAIL

08

THREE NODES: A CITY SECTION FROM TRAINING TO ADOPTION



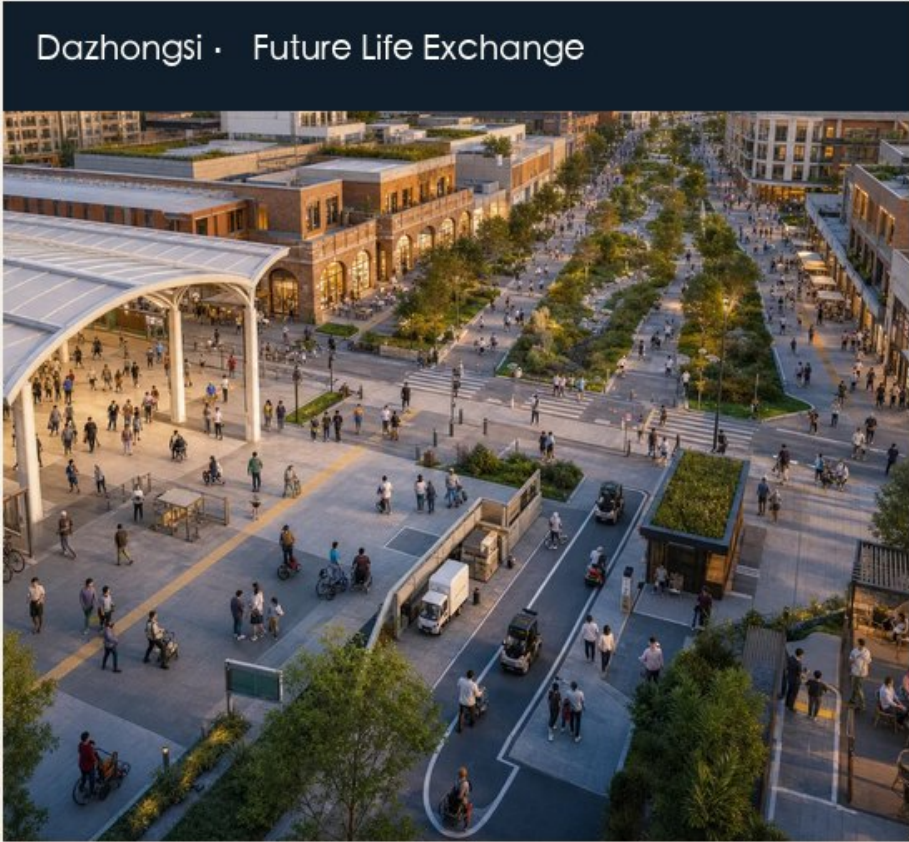
Bounded loop + depot + heat garden

Concept rendering only; ground-floor boundaries, fire, heritage and utilities require professional verification.



Model room + assembly court + human window

Concept rendering only; ground-floor boundaries, fire, heritage and utilities require professional verification.



Four quadrants + assistive tech + adoption clinic

Concept rendering only; ground-floor boundaries, fire, heritage and utilities require professional verification.

RULES FIRST, SPACE SECOND, SCALE LAST

14 PROJECTS / RACI / CAPEX + OPEX

09

RULES FIRST, SPACE SECOND, SCALE LAST

Relative resource bands: L low / M medium / H high; not an investment estimate.

PROJECT	PHASE	SPACE / GOVERNANCE DELIVERY	CAPEX	OPEX
P01	0-1	300 m Centennial Resonance Spine pilot	M	M
P02	0-1	Open semantic twin base	M	M
P03	1	Physical AI safety training loop	M	H
P04	1	World-model evaluation and red-team lab	M	M
P05	1-2	Site heat-reuse study and winter garden	H	M
P06	1-2	Civic Intelligence Academy	M	M
P07	1	Human Service Court	L	H
P08	1-2	Open Contribution Archive	L	M
P09	2	Dazhongsi four-quadrant ground reconnection	H	M
P10	1-2	Assistive Technology Library	M	M
P11	1-2	Quiet lane and accessible mobility network	M	M
P12	1-2	Blue-green cooling and rainwater nodes	M	L
P13	0	City AI Rights Charter and independent audit desk	L	M
P14	2-3	Global Future City Season	L	M

Four gates: official geometry / ownership + fire / transport + utilities / rights, energy + independent review. Any failed gate means stop, modify or withdraw.

THE TEMPERATURE OF TECHNOLOGY

HEAT / WATER / CARE

THE TEMPERATURE OF TECHNOLOGY

010



The winter garden is an energy feasibility test, not a promise

Put servers, heat pumps, thermal storage, rainwater and public comfort on one energy ledger. Disclose cost per kWh, per square metre and per season, with a conventional fallback when the test fails.

Night public life: technology recedes; human encounters stay foregrounded

SIXTEEN EVERYDAY PROTOTYPES

N01

World Model Sandbox

Visible purpose / minimum data / human review / exit
/ retrospective

N02

Physical AI Real-scene Training Loop

Visible purpose / minimum data / human review / exit
/ retrospective

N03

Robot Safety Certification Street

Visible purpose / minimum data / human review / exit
/ retrospective

N04

Compute-to-Climate Winter Garden

Visible purpose / minimum data / human review / exit
/ retrospective

N05

Municipal Robot Depot

Visible purpose / minimum data / human review / exit
/ retrospective

C01

Open Semantic Twin Room

Visible purpose / minimum data / human review / exit
/ retrospective

C02

Civic AI Assembly

Visible purpose / minimum data / human review / exit
/ retrospective

C03

Human Service Court

Visible purpose / minimum data / human review / exit
/ retrospective

SIXTEEN EVERYDAY PROTOTYPES

C04

Inclusive Co-test Workshop

Visible purpose / minimum data / human review / exit
/ retrospective

C05

Open Contribution Archive

Visible purpose / minimum data / human review / exit
/ retrospective

C06

Urban Skills Academy

Visible purpose / minimum data / human review / exit
/ retrospective

S01

Assistive Technology Library

Visible purpose / minimum data / human review / exit
/ retrospective

S02

Spatial Computing Everyday
Street

Visible purpose / minimum data / human review / exit
/ retrospective

S03

Smart Device Adoption Clinic

Visible purpose / minimum data / human review / exit
/ retrospective

S04

Four-quadrant Transit Hall

Visible purpose / minimum data / human review / exit
/ retrospective

S05

Global Future City Season

Visible purpose / minimum data / human review / exit
/ retrospective

FOURTEEN RENEWAL PROJECTS

P01 300 m Centennial Resonance Spine pilot	P02 Open semantic twin base	P03 Physical AI safety training loop	P04 World-model evaluation and red-team lab
P05 Site heat-reuse study and winter garden	P06 Civic Intelligence Academy	P07 Human Service Court	P08 Open Contribution Archive
P09 Dazhongsi four-quadrant ground reconnection	P10 Assistive Technology Library	P11 Quiet lane and accessible mobility network	P12 Blue-green cooling and rainwater nodes
P13 City AI Rights Charter and independent audit desk	P14 Global Future City Season		

UNKNOWN STAYS UNKNOWN

- Provisional rough geometry is not an official redline; recalculate the full system when official geometry arrives.
- Building prototypes are not a survey, demolition list or statutory development quantum.
- AI scenarios retain human service, appeal, off and non-digital alternatives.
- Global cases inform text methods only; no third-party visual asset is copied.

This open co-creation proposal does not replace statutory planning or constitute an approved government decision.